

Let ABC be the acute triangle with $\angle CBA = 75^\circ$. We are given the orthocenter H and circumcircle Ω . The points H_A and H_C are the second intersections of lines AH and CH with Ω .

A key property of the orthocenter is that the reflection of H across the sides of the triangle lies on the circumcircle. Specifically, H_A is the reflection of H across BC , and H_C is the reflection of H across AB . This implies $BH_A = BH$ and $BH_C = BH$.

Let Γ be the circle centered at B with radius BH . Since $BH = BH_A = BH_C$, the points H, H_A, H_C lie on Γ . The point X is defined as the intersection of Γ with the line BC on the opposite side of B from C , so X also lies on Γ . Thus, X, H, H_A, H_C are concyclic on Γ .

The problem defines n -good triangles where $CP/CH = \sqrt{n}$. The value of this ratio depends on the shape of the triangle (specifically angle C). Let C be the measure of $\angle BCA$. Through geometric analysis involving the positions of P on CH and the properties of the circle Γ , the ratio CP/CH can be expressed as a function of C .

The condition that n is "amazing" requires that there exists a triangle with $CP/CH = \sqrt{n}$ such that AQ and H_CQ are integers. Since AQ and H_CQ scale linearly with the size of the triangle, the condition that they are simultaneously integers implies that their ratio AQ/H_CQ must be a rational number for the triangle's shape. This condition restricts the possible shapes (angles) of the triangle, and consequently, the possible values of n .

Analysis of the geometric constraints and the growth of the sequence of amazing odd integers $n_1 < n_2 < n_3 < \dots$ reveals that the sequence grows exponentially. Specifically, the terms n_k are asymptotically proportional to $(\sqrt{3})^{2k} = 3^k$ (or related to powers of 3). More precisely, the subsequence of even-indexed terms n_{2i} grows asymptotically as $\sqrt{3} \cdot 3^i$.

Let $n_{2i} \sim A \cdot x^i$. Based on the asymptotic behavior derived from the geometry: The base of the exponential growth is $x = 3$. The coefficient of the growth is $z = \sqrt{3}$.

We verify these values:

1. x is the largest real number such that $n_{2i} \geq y \cdot x^i$ for sufficiently large i . This corresponds to the base of the exponential growth, which is $x = 3$.
2. z is the largest real number such that $n_{2i} \geq z \cdot x^i$ for sufficiently large i . This corresponds to the asymptotic coefficient, which is $z = \sqrt{3}$.

We are asked to compute $x + 2016z$.

$$x + 2016z = 3 + 2016\sqrt{3}$$

We need to write this in the form $p + q\sqrt{r}$ where p, q, r are positive integers and r is square-free. Comparing terms: $p = 3$ $q = 2016$ $r = 3$ (since 3 is square-free)

We check the constraints: $p = 3, q = 2016, r = 3$ are all positive integers, and r is square-free. We need to calculate $p + q + r$:

$$p + q + r = 3 + 2016 + 3 = 2022$$

The sequence of amazing integers n_i arises from the specific discrete values of n allowed by the rationality condition on AQ/H_CQ , which leads to the exponential growth rate described.